

Radioisotope therapy for advanced prostate cancer

Radioisotope therapy can be used to treat prostate cancer that has spread to the bones. It is not suitable for prostate cancer that has spread to other areas.

On this page

- [What is radioisotope therapy?](#)
- [How radioisotope therapy is given](#)
- [After radioisotope therapy](#)
- [Side effects of radioisotope therapy](#)
- [Get in touch](#)

What is radioisotope therapy?

You may have a radioisotope if [hormonal therapy](#) (ADT) is no longer helping to control prostate cancer in the bones. It may be given after chemotherapy treatment.

This treatment uses a radioactive substance called a radioisotope which is injected into a vein.

Radioisotopes are more easily absorbed by cancer cells than normal cells. This means cancer cells get a higher dose of radiation while normal cells are protected. All areas of bone affected by cancer will be treated. Radioisotope therapy can be helpful when cancer affects many areas of bone.

How radioisotope therapy is given

The radioisotope used is called radium-223 (Xofigo®). You usually have it once every 4 weeks, for 6 treatments. You have radioisotope treatment as an outpatient in the nuclear medicine department of a hospital. You go home on the same day.

Before radioisotope therapy, you have a [blood test](#) to make sure your blood cells are at a normal level. This is because radioisotopes can reduce the number of blood cells.

The radiographer gives you the radioisotope as an injection into a vein in the arm. This only takes a few minutes. When they are finished, they cover the area with a small plaster.

After radioisotope therapy

After the injection, you will have some radioisotopes in your stools (poo), urine (pee) and blood. This means you need to do things to protect yourself and others from radiation. You only need to take precautions for 1 week.

When you go to the toilet, your cancer team will advise you to sit down to use the toilet rather than standing or using a urinal. You will be asked to flush the toilet twice with the lid down. You will also need to wear gloves if you need to clean up bodily fluids.

Your cancer team can also talk to you about having sex after treatment. They will advise you about any precautions to take.

It is also important to prevent making someone pregnant after radioisotope therapy and for at least 6 months after treatment finishes. Your cancer team will advise you about what types of contraception to use and for how long.

It is usually still safe for you to be in close contact with other people, including children. Your cancer team will explain more.

The cancer team will explain more about protecting others from radiation.

Side effects of radioisotope therapy

The side effects of radioisotope therapy are usually mild. You may have some diarrhoea or feel slightly sick. Your doctor can give you drugs to control these effects if needed. It can also make you feel tired for a while. Sometimes it can cause a flare (worsening) in symptoms such as pain, but this usually improves quite quickly.

The treatment can sometimes reduce the number of blood cells for a short time. This is not usually a problem. But if you notice any unusual bruising or bleeding, contact the hospital straight away on the 24-hour number you have been given.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our prostate cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

Prostate cancer: diagnosis and management. NICE guideline [NG131] Published: 09 May 2019 Last updated: 15 December 2021 (accessed October 2024) <https://www.nice.org.uk/guidance/ng131>

Parker, C. et al. Prostate cancer: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. Annals of Oncology, Volume 31, Issue 9, 1119–1134 (accessed October 2024) [https://www.annalsofoncology.org/article/S0923-7534\(20\)39898-7/fulltext#articleInformation](https://www.annalsofoncology.org/article/S0923-7534(20)39898-7/fulltext#articleInformation)

Cancer Information Development team

Author

Dr Ursula McGovern

Reviewer

Consultant Medical Oncologist & Honorary Associate Professor
University College Hospitals, London

Date reviewed

Reviewed: 01 February 2025 | Next review: 01 February 2028

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Get in touch



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email us

Email the Macmillan Fundraising team if you have any questions.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.